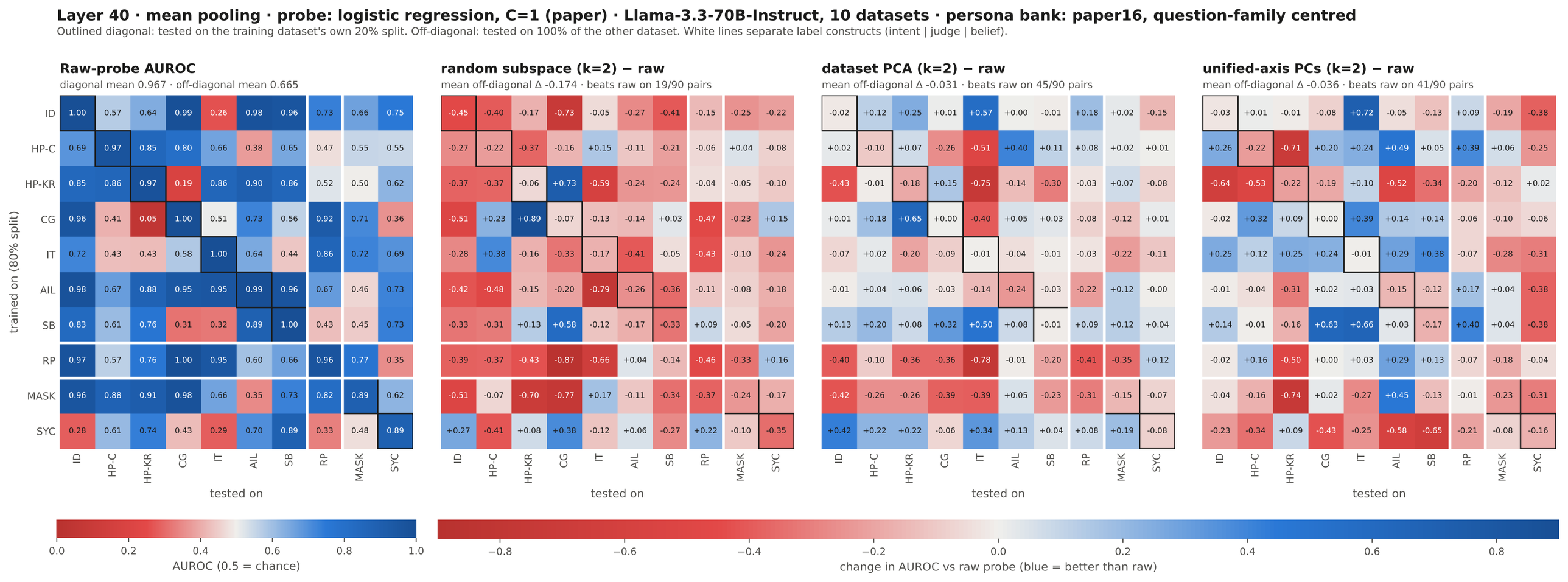

Layer 40 · mean pooling · probe: logistic regression, C=1 (paper) · Llama-3.3-70B-Instruct, 10 datasets · persona bank: paper16, question-family centred
Outlined diagonal: tested on the training dataset's own 20% split. Off-diagonal: tested on 100% of the other dataset. White lines separate label constructs (intent | judge | belief).
Raw-probe AUROC
diagonal mean 0.967 · off-diagonal mean 0.665
random subspace (k=2) − raw
mean off-diagonal Δ -0.174 · beats raw on 19/90 pairs
dataset PCA (k=2) − raw
mean off-diagonal Δ -0.031 · beats raw on 45/90 pairs
unified-axis PCs (k=2) − raw
mean off-diagonal Δ -0.036 · beats raw on 41/90 pairs
trained on (80% split)
ID HP-C HP-KR CG IT AIL SB RP MASK SYC
tested on
AUROC (0.5 = chance)
change in AUROC vs raw probe (blue = better than raw)